

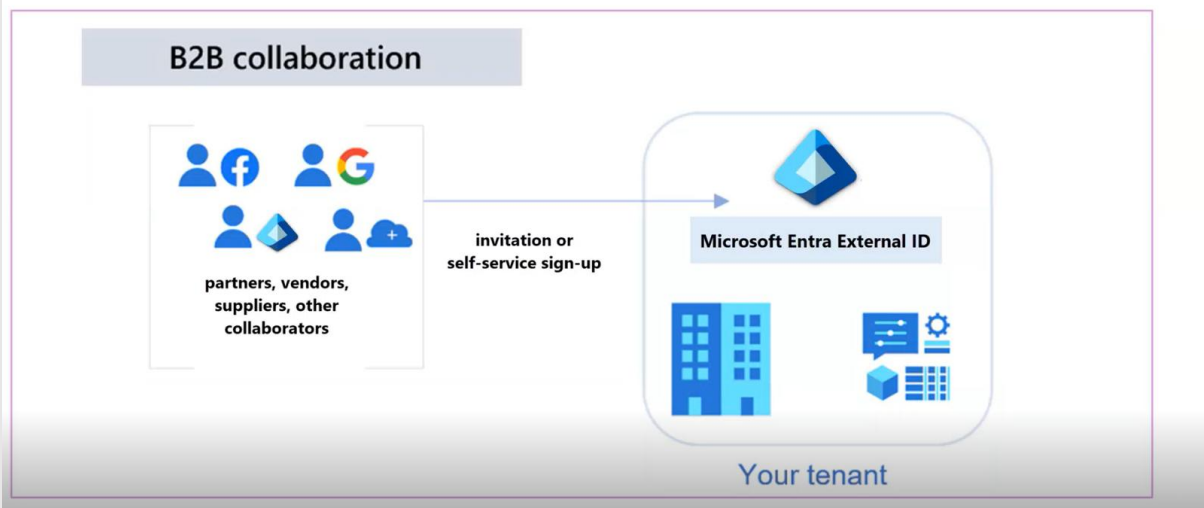
Cloud Computing 101 – Part 5

Identity, Access and Security in Microsoft Azure

Identity Management

Not every person who needs access to your company's systems is an employee. Partners, vendors and contractors exist outside your organization but still need to collaborate on shared resources. Microsoft Entra External ID B2B solves this elegantly. It allows those outside users to bring their own identity, whether a Microsoft account, a Google account or any other provider, and gain controlled access into your tenant without you needing to create and manage separate accounts for them. Think of it as giving a trusted visitor a temporary access card to specific floors of your building, rather than issuing them a full staff ID. The two entry points available, which are invitation and self-service sign-up, give organizations flexibility in how they onboard these external collaborators.

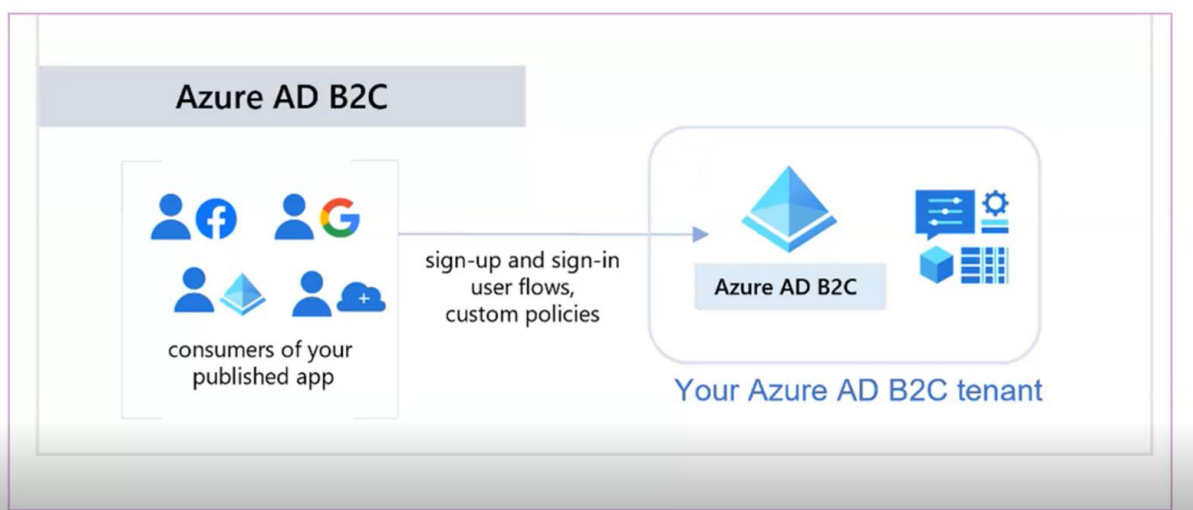
Microsoft Entra External ID B2B



Customer Identity

While B2B handles business to business collaboration, B2C is built for an entirely different audience: the end customers of your published application. If you are a developer or a business building a consumer-facing app such as a mobile banking app, an e-commerce platform or a healthcare portal, you need a way for millions of customers to register, log in and manage their own accounts without any involvement from your IT team. Azure AD B2C handles all of this. It provides a dedicated tenant with fully customizable sign-up and sign-in flows, custom policies for complex authentication scenarios and support for social identity providers. Your customers never even know they are touching Microsoft infrastructure because the entire experience is branded as yours.

Azure AD External Identities B2C



Access Policy

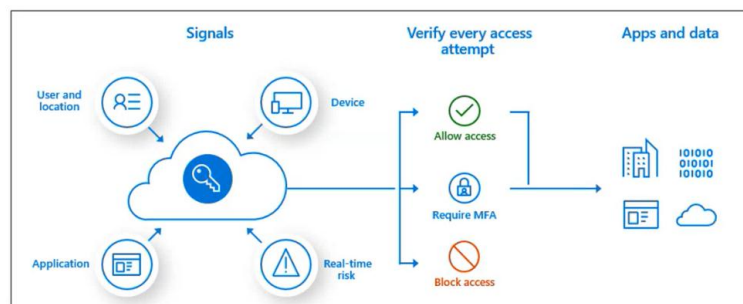
Conditional Access

Passwords alone are no longer enough to protect organizational resources. Conditional Access is Microsoft's answer to this challenge. Instead of a simple allow or deny gate at login, it evaluates multiple real-time signals simultaneously and then decides whether to grant access, block it outright or step up the verification by requiring Multi-Factor Authentication. Those signals include who you are, what device you are on, where you are connecting from, which application you are trying to reach and whether any risk has been detected in your session. This makes it possible, for example, to let an employee log in freely from their managed corporate laptop while requiring extra verification for the same employee on a personal phone, all automatically and with no help desk ticket required. It is the intelligence layer that sits between your users and your data.

Conditional Access

Conditional Access is used to bring signals together, to make decisions, and enforce organizational policies.

- User or group membership
- IP location
- Device
- Application
- Risk detection



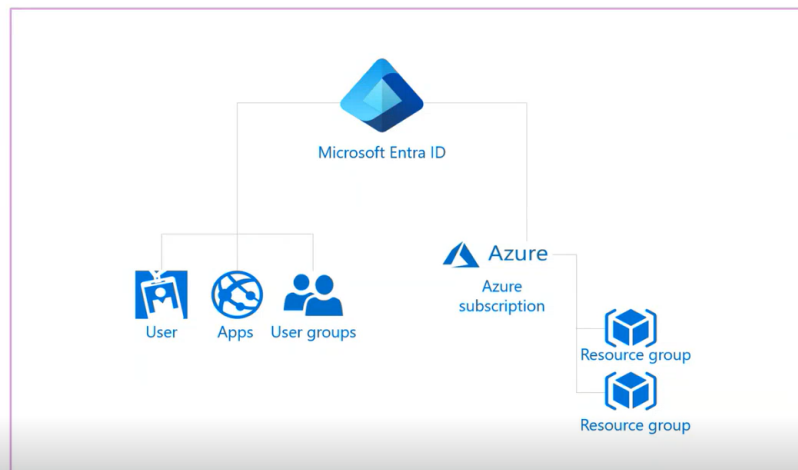
Permissions Management

Role-Based Access Control

In any organization, not everyone should have access to everything. A junior developer should not have the ability to delete production databases. A finance analyst should be able to view cost reports but not reconfigure virtual machines. Role-Based Access Control in Azure enforces this principle of least privilege at scale. Rather than assigning permissions directly to individual users, which becomes unmanageable as teams grow, it works by assigning roles to users, applications or groups. Those roles carry a predefined set of permissions. When someone's responsibilities change, you simply update their role assignment. What makes Azure RBAC especially powerful is that it operates across the entire resource hierarchy, from the subscription level all the way down to a single storage container, giving organizations precise control over who can do what and where.

Role-based access control

- Fine-grained access management.
- Segregate duties within the team and grant only the amount of access to users that they need to perform their jobs.
- Enables access to the Azure portal and controlling access to resources.



Security Philosophy

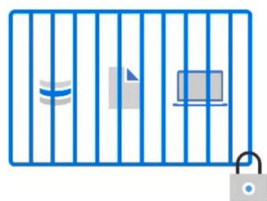
Zero Trust

For decades, the dominant security model was built around a hard outer shell called the network perimeter. If you were inside the network, you were trusted. If you were outside, you were not. This worked reasonably well when everyone worked from the office on company-managed devices. Today, with remote work, cloud services, personal devices and third-party integrations, that perimeter no longer exists in any meaningful way. Zero Trust is the modern security philosophy that replaces the old model with one simple principle: never trust and always verify. It assumes that threats can exist both outside and inside the network, so every access request regardless of where it comes from must be authenticated, authorized and continuously validated. The diagram captures this shift clearly, moving from locking everything inside a fence to protecting each asset individually wherever it lives, backed by a central policy engine.

Zero Trust

Secure assets where they are with Zero Trust

Simplify security and make it more effective



Classic Approach

Restrict everything to a 'secure' network



Zero Trust

Protect assets anywhere with central policy

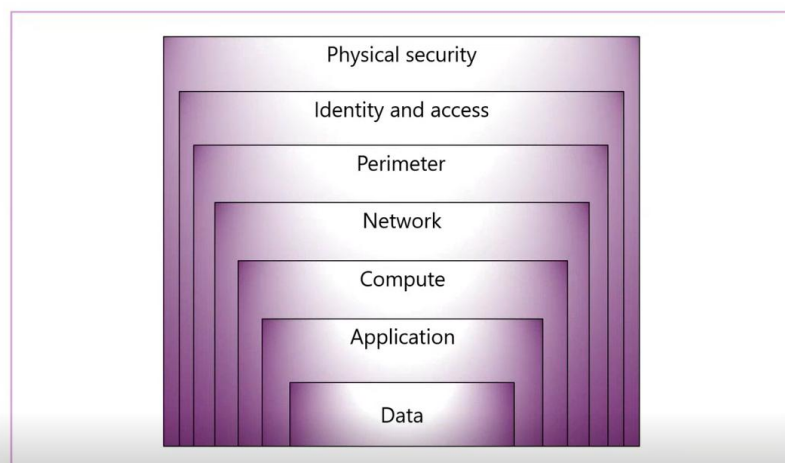
Security Architecture

Defense in Depth

If Zero Trust is the philosophy, Defense in Depth is the architecture that brings it to life. The core idea is straightforward: do not rely on a single security control. Instead, build security in concentric layers so that if one layer is breached, the next one is already in place to slow down or stop an attacker. The seven layers shown in the diagram each serve a distinct purpose. Physical security prevents unauthorized entry to hardware. Identity and access controls ensure only the right people get through. The network layer limits how far an attacker can move once inside. Compute security hardens the operating environment. Application security removes vulnerabilities at the code level. And at the very centre, the data itself is encrypted and tightly controlled. An attacker who bypasses one layer immediately faces the next. This is why cloud security professionals think in layers and not walls.

Defense in depth

- A layered approach to securing computer systems.
- Provides multiple levels of protection.
- Attacks against one layer are isolated from subsequent layers.



Threat Protection

Microsoft Defender for Cloud

Knowing your security architecture is sound in theory is one thing. Knowing that it is holding up in practice, right now, across hundreds of live resources is another thing entirely.

Microsoft Defender for Cloud is the visibility and threat protection layer that answers the practical question of whether your environment is actually secure. It continuously monitors your Azure resources and on-premises servers, scores your overall security posture, surfaces actionable recommendations and actively detects threats in real time. The dashboard shown in class is not just a reporting tool. It is an operational command centre where you can see your secure score, track regulatory compliance, review active threats, manage firewall rules and drill into specific vulnerable workloads all from a single place. For anyone going into cloud security or cloud administration, getting hands-on with this tool early is one of the most valuable things you can do.

Microsoft Defender for Cloud

Microsoft Defender for Cloud is a monitoring service that provides threat protection across both Azure and on-premises datacenters.

- Provides security recommendations.
- Detect and block malware.
- Analyze and identify potential attacks.
- Just-in-time access control for ports.

